# Hyperbolas

## Standard form

1 For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find:

a the vertices

b the foci

c the equations of the asymptotes

2 Write the standard-form equation of the hyperbola with vertices $(0, \pm 2)$ and foci $(0, \pm 3)$.

3 Consider the equation $4x^2 - y^2 + 8x + 4y - 4 = 0$.

a Write the equation in standard form by completing the square.

b Identify the conic and state its center.

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## Identifying conics

4 Identify each conic section from its equation:

a $x^2 + y^2 = 49$

b $\frac{x^2}{9} + \frac{y^2}{25} = 1$

c $y^2 - 4x^2 = 16$

d $y = 3x^2 - 6x + 1$

5 A hyperbola has asymptotes $y = \pm 2x$ and vertices $(\pm 3, 0)$. Find its equation.

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## Hyperbolas centered away from the origin

6 For the hyperbola $\frac{(y+1)^2}{4} - \frac{(x-3)^2}{9} = 1$, find:

a the center

b the vertices

c the direction of opening

7 Complete the square to write $9x^2 - 4y^2 - 18x - 16y - 43 = 0$ in standard form, and state the center.

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## Eccentricity and applications

8 Find the eccentricity of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, to two decimal places, and state whether it is closer to a pair of straight lines or to a circle.

9 A ship uses two radio towers 200 km apart (foci) to navigate via hyperbolic positioning: the difference in distances from the ship to each tower is constant at 120 km. Write the equation of the hyperbola (centered at the midpoint of the towers, opening left-right), in standard form.